a Feature extraction

Whole slide image

Tiling

N tiles

Encoding

Tile encoder

❄: frozen

Feature matrix $\boldsymbol{X}$

$\boldsymbol{X} \in \mathbb{R}^{N \times D}$

D

N: number of tiles
D: feature dimension

b Attention-based MIL

Attention

Weighted aggregation

Slide representation

Slide-level classifier

Prediction $\hat{y}$

Low High

⑦ Prediction shift

Source → target

Prediction $\hat{y}'$

Attention-based MIL

Tile encoder ❄

⑥ Progressive slide-level counterfactual

1 2 3 4 5 6 7 8 9

Attention-guided region selection

c Concept-guided counterfactual generation

① Concept discovery

Sparse autoencoder (SAE)

SAE encoder

SAE latent $\boldsymbol{z}$

SAE decoder

Reconstructed feature

② Concept bank

Concept C_1

Concept C_2

Concept C_3

Concept C_L

Top-k representative tiles per concept

③ Concept selection

Concept–class association *(SAE activations)*

Class A
Class B

$C_1 C_2 C_3$ ... $\boldsymbol{C_{j*}}$ ... C_L

High Low

④ Concept steering

Original feature $\boldsymbol{x}$

SAE encoder

SAE latent $\boldsymbol{z}$

Steered latent $\boldsymbol{z}'$

Prototype $\boldsymbol{p_{j*}}$ derived from target concept $\boldsymbol{C_{j*}}$

$\boldsymbol{z}' = \boldsymbol{z} + s(\boldsymbol{p_{j*}} - \boldsymbol{z})$

SAE decoder

Steered feature $\boldsymbol{x}'$

s: steering strength

⑤ Region-level counterfactual generation

Original image

Conditioned on steered feature $\boldsymbol{x}'$

Counterfactual image ★

Diffusion model ❄